

Diabetic Retinopathy Severity Classification Using Deep Learning with Ordinal Regression

Foundations of Artificial Intelligence NEU CS5100

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1. Introduction

Diabetic Retinopathy (DR) is one of the leading causes of preventable blindness worldwide and affects nearly one-third of diabetic patients. DR develops when prolonged hyperglycemia damages retinal blood vessels, causing them to swell, leak, and form characteristic lesions visible in fundus photographs. Early detection is critical as timely intervention can significantly reduce the risk of vision loss. However, manual grading by ophthalmologists is time-consuming and subject to variation between observers, which automated classification systems can aid.

Recent advances in deep learning, particularly convolutional neural networks (CNNs), show strong potential for automated DR detection and classification. CNN architectures typically consist of convolutional layers for feature extraction, pooling layers for dimensionality reduction, and fully connected layers for classification. Transfer learning strategies, which leverage models pretrained on large image datasets like ImageNet, have proven especially effective for medical imaging tasks where labeled data is limited.

DR severity is defined by the presence and progression of retinal lesions. According to the international clinical DR scale summarized in Alyoubi et al., five levels are recognized: No DR (Class 0), Mild non-proliferative DR with microaneurysms only (Class 1), Moderate non-proliferative DR (Class 2), Severe non-proliferative DR (Class 3), and Proliferative DR (Class 4). Because treatment decisions depend on precise staging, accurate multi-level classification is clinically essential.

A key finding from the literature is that most studies focus on binary classification (DR vs. no DR) rather than full severity grading. Alyoubi et al. report that approximately 73% of reviewed studies perform binary classification, while only 27% attempt multi-stage severity prediction. Binary systems often achieve high reported accuracy exceeding 90%, but fail to distinguish between mild, moderate, severe, and proliferative stages, information critical for planning DR treatment.

The literature also identifies a limitation in how DR classification systems are evaluated. Performance is typically measured using accuracy, sensitivity, specificity, and area under the curve (AUC). However, these metrics treat classes as categorical (nominal) rather than ordinal (ranked), which is problematic when predicting severity stages that have a natural order. A

system that predicts Class 2 when the true label is Class 3 has made a smaller clinical error than one that predicts Class 0, yet standard metrics penalize both equally.

This project addresses two gaps identified in the literature. First, it implements a full five-stage severity classifier rather than binary detection. Second, it explores ordinal regression as an alternative to standard categorical classification, directly addressing the ordinal nature of DR severity levels. The project compares multiple approaches including transfer learning with ResNet-50 and EfficientNet-B3, class-weighted loss functions to handle dataset imbalance, and ordinal regression loss to respect the ordered nature of severity stages.

2. Methods

Dataset

This project uses the APTOS 2019 Blindness Detection dataset from Kaggle, which contains retinal fundus photographs labeled with DR severity grades from 0 (no DR) to 4 (proliferative DR). The dataset includes 2,930 training images and 366 validation images. The dataset exhibits significant class imbalance: Class 0 (No DR) contains approximately 1,430 images (49%), while Class 3 (Severe) has only about 150 images (5%). This imbalance reflects real-world clinical distributions but presents challenges for model training.



Preprocessing Pipeline

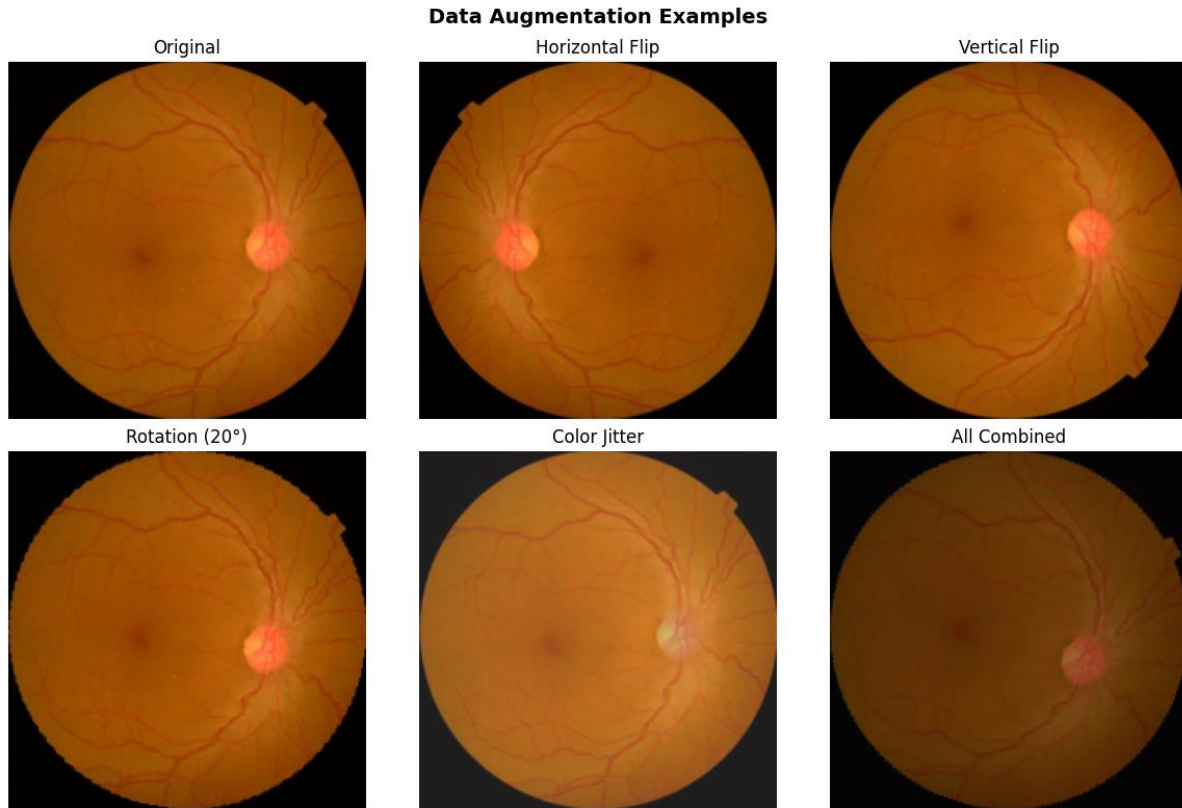
All images were preprocessed using a consistent pipeline implemented in PyTorch. Images were resized to 224×224 pixels, the standard input size for ResNet and EfficientNet architectures.

Pixel values were normalized using ImageNet statistics (mean = [0.485, 0.456, 0.406], standard deviation = [0.229, 0.224, 0.225]) to match the distribution the pretrained models expect.

Data augmentation was applied during training to reduce overfitting and improve generalization.

Augmentation transforms included random horizontal flips, random vertical flips, random

rotation up to 20 degrees, and color jittering with brightness and contrast variations of 20%. These augmentations are particularly appropriate for fundus images, which have no inherent orientation and can exhibit natural variation in lighting conditions.



The validation set used only resizing and normalization without augmentation to ensure consistent evaluation.

Model Architectures

Four model variants were trained and evaluated:

ResNet-50 Baseline: A ResNet-50 model pretrained on ImageNet with the final fully connected layer replaced to output 5 classes. This served as the baseline for comparison.

EfficientNet-B3: An EfficientNet-B3 model pretrained on ImageNet with the classifier layer modified for 5-class output. EfficientNet uses compound scaling to balance network depth, width, and resolution, often achieving better accuracy with fewer parameters than ResNet.

ResNet-50 with Weighted Loss: The same ResNet-50 architecture trained with class-weighted cross-entropy loss. Weights were calculated as the inverse of class frequency, causing the model to penalize errors on minority classes more heavily. This approach directly addresses class imbalance.

ResNet-50 with Ordinal Regression: A ResNet-50 model modified to output 4 values instead of 5, trained with ordinal regression loss. Instead of predicting which class an image belongs to, ordinal regression predicts whether the severity exceeds each threshold: Is severity ≥ 1 ? Is severity ≥ 2 ? And so on. The final class prediction is determined by counting how many thresholds the model predicts are crossed. This approach respects the ordered nature of severity levels.

Training Configuration

All models were trained for 5 epochs using the Adam optimizer with a learning rate of 0.0001. Batch size was set to 32. Training was performed on an NVIDIA RTX 4000-series GPU with CUDA acceleration. Each model took approximately 12-15 minutes to train.

Evaluation Metrics

Models were evaluated using multiple metrics:

Validation Accuracy: The percentage of correctly classified images.

Quadratic Weighted Kappa: The standard metric for ordinal DR classification, which accounts for the ordered nature of severity grades. Kappa penalizes predictions that are far from the true label more than those that are close, making it more clinically meaningful than accuracy alone.

Per-Class Accuracy: The accuracy for each individual severity class, revealing how well models perform on minority classes.

Confusion Matrices: Visual representations of prediction patterns showing which classes are commonly confused.

3. Results

Overall Performance

Table 1 summarizes the performance of all four model variants on the validation set.

Model	Val Accuracy	Quadratic Kappa
ResNet-50 Baseline	82.79%	0.8876
EfficientNet-B3	82.79%	0.8948
ResNet-50 Weighted	79.78%	0.8688
ResNet-50 Ordinal	77.87%	0.8868

The ResNet-50 baseline and EfficientNet-B3 achieved identical validation accuracy of 82.79%. However, EfficientNet-B3 achieved a higher quadratic weighted kappa of 0.8948 compared to 0.8876 for ResNet-50, indicating that when EfficientNet makes errors, they tend to be smaller in magnitude (predicting adjacent classes rather than distant ones).

The weighted loss approach reduced overall accuracy to 79.78% with a kappa of 0.8688. The ordinal regression model achieved 77.87% accuracy with a kappa of 0.8868. While these overall numbers appear lower, per-class analysis reveals a more nuanced picture.

Per-Class Performance

Table 2 shows the accuracy for each severity class across all models.

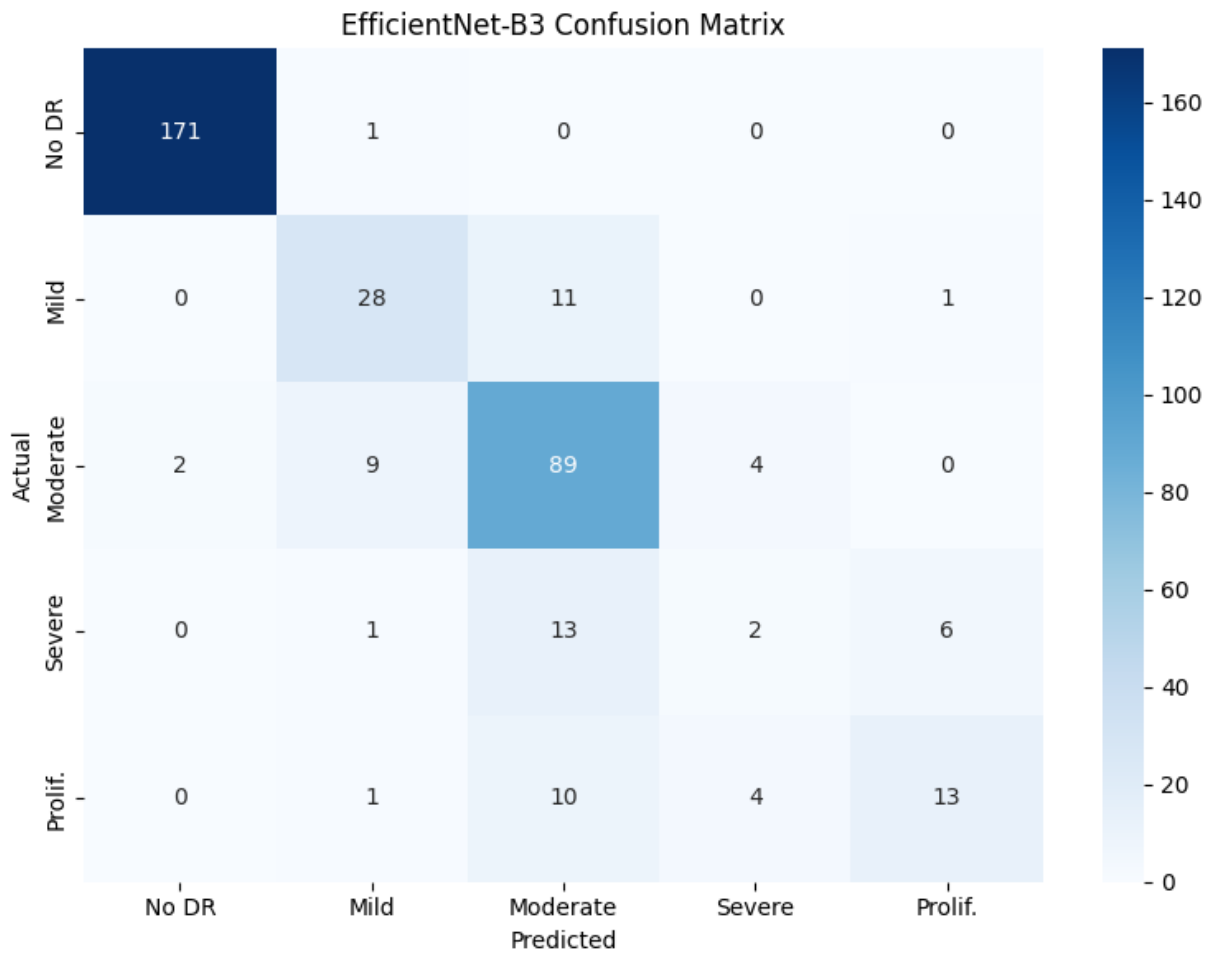
Model	Class 0 (No DR)	Class 1 (Mild)	Class 2 (Moderate)	Class 3 (Severe)	Class 4 (Prolif.)
ResNet-50 Baseline	99.4%	72.5%	76.0%	40.9%	53.6%
EfficientNet-B3	99.4%	70.0%	85.6%	9.1%	46.4%
ResNet-50 Weighted	98.8%	77.5%	63.5%	40.9%	57.1%
ResNet-50 Ordinal	99.4%	40.0%	67.3%	40.9%	67.9%

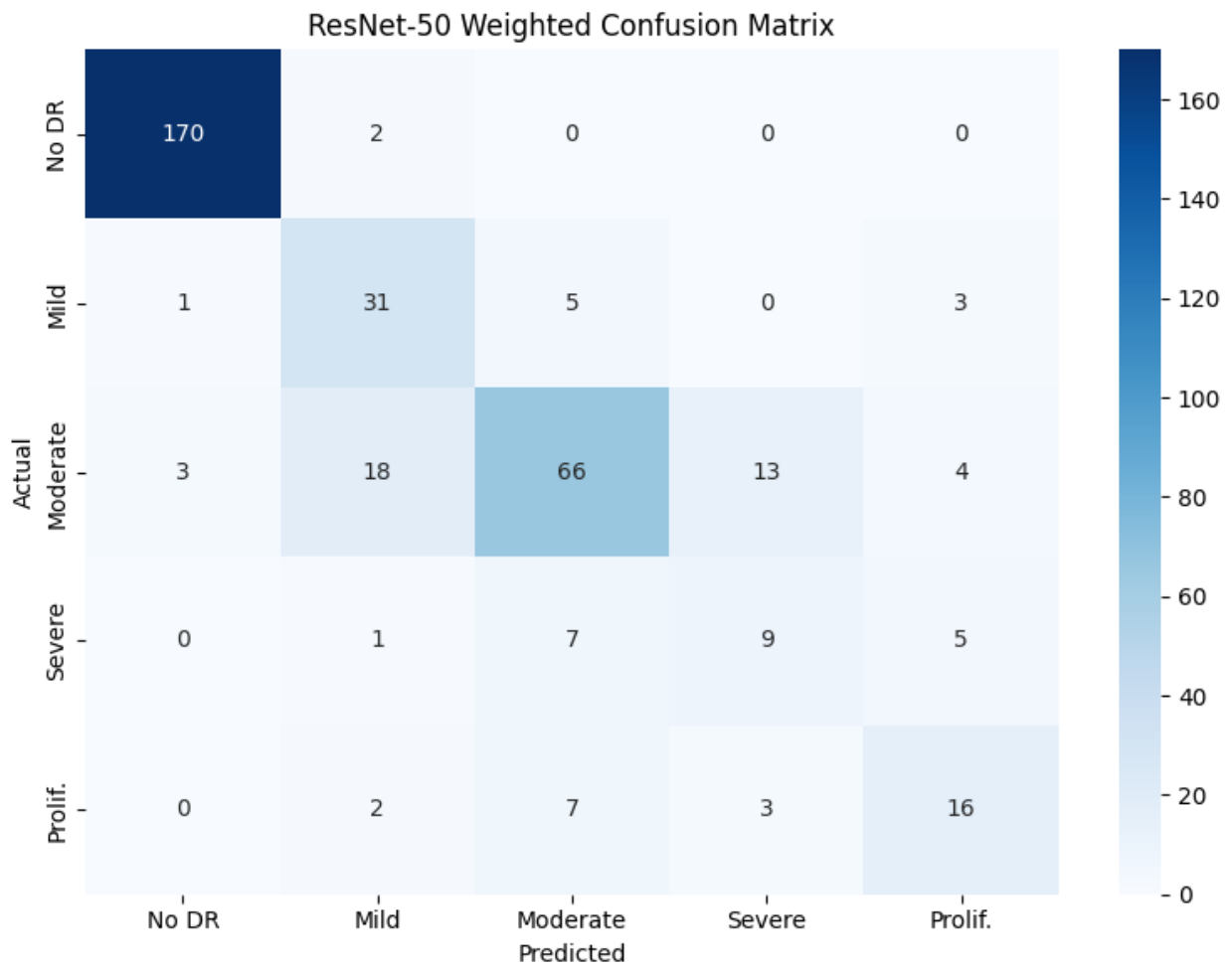
All models achieved excellent performance on Class 0 (No DR), the majority class, with accuracies above 98%. This is expected given that nearly half the training data belongs to this class.

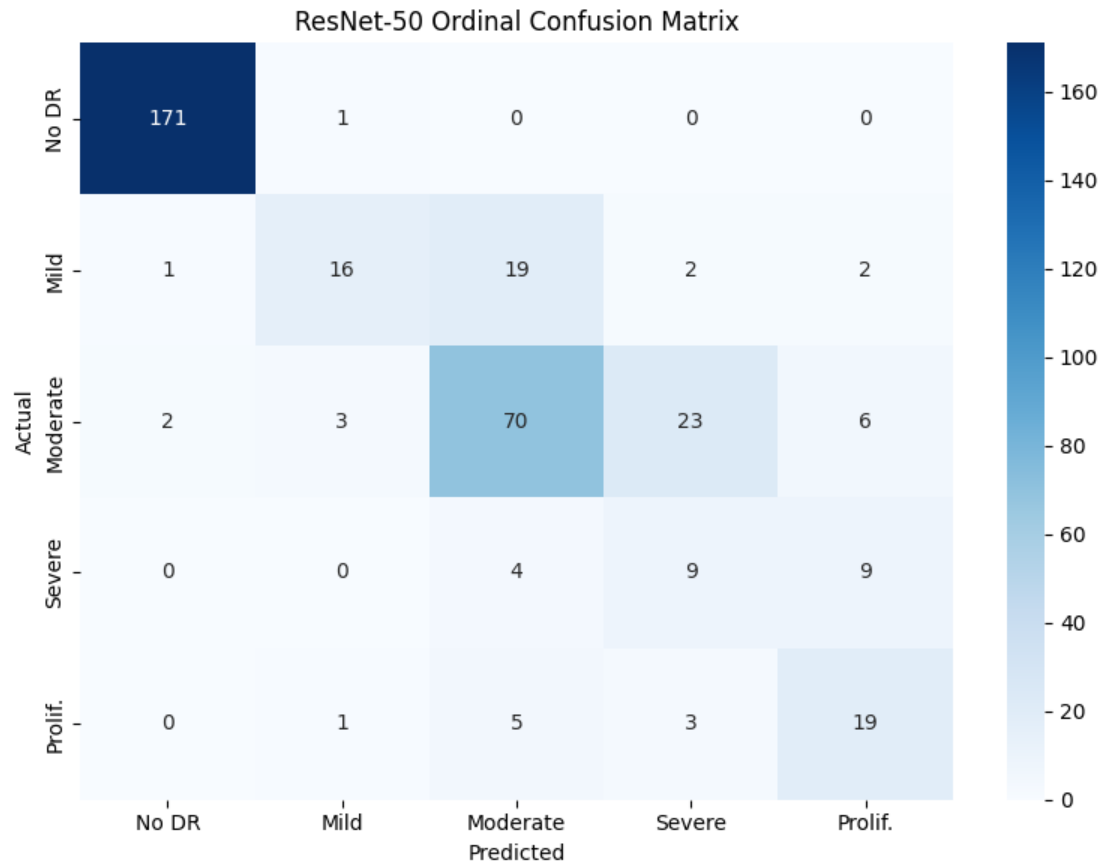
Class 3 (Severe) proved most challenging for all models. EfficientNet-B3 performed particularly poorly on this class with only 9.1% accuracy, despite achieving the highest overall kappa. The ResNet variants all achieved 40.9% on Class 3.

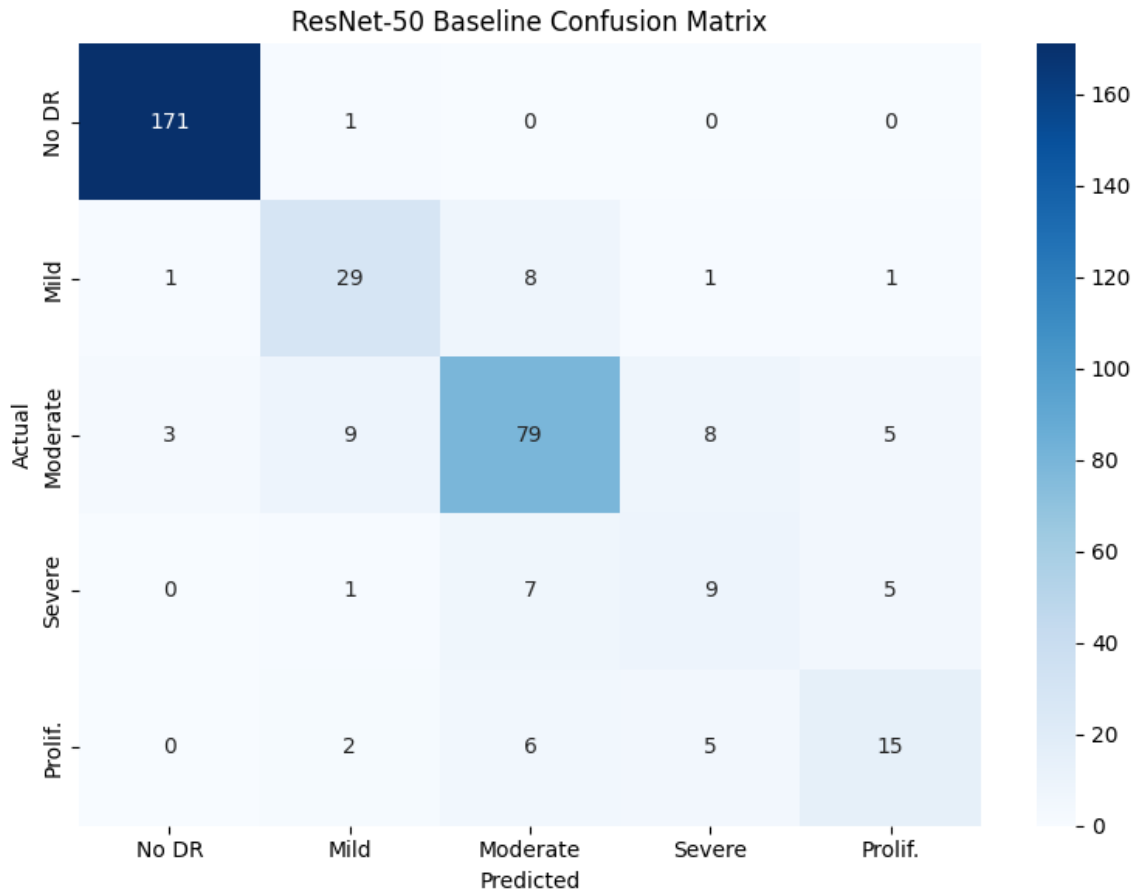
The most notable finding is that the ordinal regression model achieved the highest accuracy on Class 4 (Proliferative DR) at 67.9%, substantially outperforming all other models. The next best was the weighted loss model at 57.1%, followed by the baseline at 53.6% and EfficientNet at 46.4%.

The weighted loss approach successfully improved Class 1 (Mild) accuracy from 72.5% to 77.5% and Class 4 accuracy from 53.6% to 57.1%, demonstrating that class weighting can improve minority class performance.









Analysis of Results

The ordinal regression model's superior performance on Class 4 (Proliferative DR) is clinically significant. Proliferative DR is the most severe stage, where abnormal blood vessel growth can lead to retinal detachment and blindness. Accurate identification of this stage is critical for urgent treatment referral. The ordinal approach's 67.9% accuracy on this class represents a 14.3 percentage point improvement over the baseline.

The ordinal model's lower overall accuracy can be attributed to its reduced performance on Class 1 (Mild), where it achieved only 40.0% compared to 72.5% for the baseline. This tradeoff may be acceptable in clinical settings where detecting severe disease is prioritized over distinguishing between mild and no DR.

EfficientNet-B3's poor Class 3 performance (9.1%) despite high overall kappa illustrates a key limitation of aggregate metrics. The model appears to have learned to avoid predicting Class 3

almost entirely, instead assigning those images to adjacent classes. Because Class 3 is rare, this strategy minimally impacts overall accuracy while severely compromising clinical utility for severe cases.

4. Conclusions

Strengths

This project successfully implemented a five-stage DR severity classifier, addressing the literature gap where most systems perform only binary classification. Comparing four distinct approaches (baseline transfer learning, alternative architecture, class weighting, and ordinal regression) provides insight into how different strategies handle the challenges of medical image classification.

The ordinal regression approach shows meaningful contribution by achieving the highest accuracy on proliferative DR (67.9%), the most clinically urgent category, supporting the hypothesis from the literature review that treating severity as ordinal rather than categorical can improve performance on the task's most important aspects.

All models achieved quadratic weighted kappa scores above 0.86, indicating strong agreement with ground truth labels that account for the ordinal nature of the classification task.

Weaknesses

Overall accuracy remained below 85% for all models. Class 3 (Severe) performance was poor across all approaches, never exceeding 41%, suggesting

that the features distinguishing severe from moderate or proliferative DR may be subtle and require either more training data, more sophisticated architectures, or explicit lesion detection.

The ordinal regression model's poor performance on Class 1 (Mild) represents a tradeoff that may not be acceptable in all clinical contexts. A more sophisticated approach might combine ordinal and categorical losses or use a hierarchical classification scheme.

The dataset's class imbalance remains a fundamental challenge. Despite attempting weighted loss, minority class performance improved only modestly. More aggressive techniques such as synthetic oversampling (SMOTE) or generative augmentation were not explored.

Future Work

Several directions could improve this work: first, implementing Grad-CAM or similar visualization techniques would provide interpretability by showing which regions of fundus images the model attends when making predictions. This could validate that the model focuses on clinically relevant features like microaneurysms and hemorrhages.

Second, combining ordinal regression with class weighting might preserve the ordinal model's strong Class 4 performance while improving its weak Class 1 results. A hybrid loss function could balance these objectives.

Third, training for more epochs with learning rate scheduling and early stopping based on validation kappa rather than accuracy might improve results. The current 5-epoch training may not have allowed models to fully converge.

Finally, methods combining predictions from multiple models could leverage their complementary strengths - for example, using EfficientNet for moderate cases and the ordinal model for severe cases.

References

- Alyoubi, W. L., Shalash, W. M., & Abulkhair, M. F. (2020). Diabetic retinopathy detection through deep learning techniques: A review. *Informatics in Medicine Unlocked*, 20, 100377. <https://doi.org/10.1016/j.imu.2020.100377>
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